MUHAMMAD UMAR

BS Computer Science Student — Web Developer — AI Enthusiast

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SUMMARY

Computer Science undergraduate at the University of Management and Technology (UMT) with internship experience in AI/ML and Generative AI. Experienced in developing LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, semantic search solutions, and machine learning models. Hands-on experience with Python, Streamlit, embeddings, vector databases, and AI API integration. Interested in building practical AI applications that solve real-world problems.

EDUCATION

University of Management and Technology (UMT) 2023 – Present

Bachelor of Science in Computer Science (BSCS)

Currently in 7th Semester

EXPERIENCE

Generative AI Intern – Big Brains *2026*

- Completed a Generative AI internship focused on Large Language Models (LLMs), prompt engineering, embeddings, vector search, and Retrieval-Augmented Generation (RAG).
- Developed practical AI applications using Python, Streamlit, and AI APIs, applying knowledge retrieval and context-aware response generation.
- Earned an internship completion certificate with an average score of 99%.

PROJECTS

Big Brains Customer Support AI Agent *Python, Streamlit, ChromaDB, Groq API*

- Developed an AI-powered customer support agent to answer questions about Big Brains courses, internships, products, services, learning resources, and policies.
- Implemented embeddings, semantic search, and knowledge retrieval using ChromaDB and Sentence Transformers.
- Integrated an LLM through the Groq API to generate context-aware responses grounded in retrieved information.
- Implemented chat history management using SQLite and designed the agent to avoid unsupported answers when relevant information is unavailable.

Hospital RAG AI Assistant *Python, Streamlit, FAISS, Groq API*

- Built a hospital-focused AI assistant that answers questions using information retrieved from hospital PDF documents.
- Implemented PDF text extraction using PyMuPDF, document chunking, embeddings, and FAISS similarity search.
- Connected retrieved document context to an LLM to generate grounded responses from the indexed knowledge base.
- Organized hospital information into Admissions, Department, Emergency, Hospital, and Patient Safety categories.

Student Dropout Prediction System *Python, Pandas, Scikit-learn*

- Developed a machine learning model using a higher education dataset containing approximately 4,424 student records and 36 features.
- Trained a Logistic Regression model to predict whether a student may drop out or continue their studies.
- Used student academic and personal information, including first- and second-semester indicators, to identify potential dropout risk.
- Applied data preprocessing, feature scaling, and model evaluation techniques to assess predictions.

E-commerce Customer Support AI Agent *Python, Streamlit, RAG, ChromaDB, Groq API*

- Developed an AI-powered customer support assistant to answer e-commerce customer queries using a knowledge base and Retrieval-Augmented Generation (RAG).
- Used semantic search and vector embeddings to retrieve relevant information and generate context-aware responses through an integrated large language model.

WhatsApp AI Chatbot *n8n, WhatsApp, AI Chat Model*

- Built an AI-powered WhatsApp chatbot using n8n and an integrated AI chat model to process user queries.
- Designed a workflow to receive WhatsApp messages and generate automated conversational responses.

TRAINING & CERTIFICATIONS

Generative and Agentic AI Application Developer *Pak Angels, Cohort 11*

- Completed practical training covering LLMs, prompt engineering, embeddings, vector databases, RAG, function calling, AI agents, and Model Context Protocol (MCP).

- Received the Top Performer award.

Generative AI Internship Completion Certificate *Big Brains*

- Certificate issued on 14 October 2026; average score of 99%.

Cyber Security Workshop – Certificate of Participation
Allah Walay Trust (AWT) – Volunteer Certificate

VOLUNTEERING

Volunteer – Allah Walay Trust (AWT)

- Participated in community service activities and developed teamwork, communication, and collaboration skills.

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, SQL

AI/ML & Generative AI: Machine Learning, Generative AI, Large Language Models (LLMs), Prompt Engineering, RAG, Embeddings, Semantic Search, AI Chatbots, Function Calling, Model Context Protocol (MCP)

Libraries & Frameworks: Streamlit, Pandas, Scikit-learn, Sentence Transformers, FAISS, ChromaDB, PyMuPDF

Web Development: HTML5, CSS3, JavaScript, jQuery, Responsive Web Design

Tools: Git, GitHub, VS Code, Groq API

Computer Science: Data Structures, Algorithms, Object-Oriented Programming, Database Systems, Compiler Design, Computer Networks

SOFT SKILLS

Problem Solving — Teamwork — Communication — Fast Learning — Adaptability — Time Management — Leadership

AREAS OF INTEREST

Artificial Intelligence — Machine Learning — Generative AI — Large Language Models — RAG Applications — AI Agents — Web Development